Cell identities and subtype evidence

All 73,052 QC-retained nuclei are accounted for. Broad identities were reviewed with multi-gene evidence, followed by independent subclustering of cardiomyocytes, fibroblasts, endothelial, mural and immune populations. Epicardial and neural/glial populations retain broad labels; a lineage-weak group remains unresolved.

59 computational subclusters were evaluated. Numerical clusters are not automatically distinct cell types. RNA-based identities and anatomical interpretations remain provisional.

Parent	Nuclei	Subclusters
Cardiomyocyte	46,485	11
Fibroblast	14,985	11
Endothelial	5,573	15
Mural	3,804	10
Immune	1,248	12

Criteria used for identity

1. Broad identity: at least three canonical lineage markers with mean log1p(counts normalized to 10,000) >=0.1 and >=10% of nuclei having >=1 corrected count; at least two markers permits only candidate status. Competing lineage panels must themselves have at least two supported markers. Minimum standardized panel score 0.3 and separation 0.25; an unqualified shared ion-channel gene cannot override a coherent lineage panel.

2. Proposed subtype: at least two supported panel genes plus the required anchor genes listed below. Detection is defined using >=1 corrected count, not any tiny positive DecontX residual. Positive evidence and exclusion-marker expression are recorded separately. Exclusion genes contribute a score penalty/check; they are not required to be completely absent.

3. Evidence status: >=50 nuclei; >=2 libraries each contributing >=10 nuclei; <=80% from one library; positive panel score >=0.3; score margin >=0.25; mean panel enrichment >=0.1 log units versus the remainder of the parent; positive within-library contrasts in >=2 libraries. These are reproducible analysis thresholds, not universal diagnostic cutoffs. They do not establish animal-level independence.

4. Cycling is a separate state: a nucleus is cycling-marker positive when >=2 of Mki67, Top2a, Cenpf, Ccnb1, Ube2c and Cdk1 have >=1 corrected count; a subcluster is cycling-enriched when >=10% meet this definition. RNA positivity does not demonstrate completed cell division.

5. Ddc-, Slit2- and Ankrd1-enriched CM programs are reported as separate marker states, not mutually exclusive anatomical cell types. An anchor, another supported program gene, and anchor enrichment >0.1 log units are required.

Important limits

All labels remain provisional RNA-based interpretations. The conduction-associated CM candidates are not asserted to be anatomically or functionally verified Purkinje cells. CAR_3 is Etv1/Gja5-enriched with limited Cntn2 detection; CAR_9 has stronger Cntn2/Etv1 support. Kirrel3-enriched fibroblasts are not assigned valve location without spatial validation. Developmental Postn-associated remodeling is not equated with pathological fibrosis. Lyve1 alone does not establish lymphatic EC identity; the lymphatic candidate requires endothelial lineage plus Prox1/Flt4 and a broader lymphatic panel.

Two libraries per age and differential QC losses (especially P7) limit abundance and developmental inference. Candidate panels can be incomplete and neighboring programs can overlap. Low detection does not prove biological absence. Cell/nucleus proportions do not equal tissue cell proportions. Descriptive marker rankings are not replicate-aware differential-expression significance tests. Spatial and/or independent reference validation is needed before treating anatomical subtype names as established.

Subtype identity criteria

Cardiomyocyte

Proposed identity	Positive marker panel	Required anchors
Atrial-like	Myl4, Myl7, Kcnj3, Sln, Nr2f2, Tbx5	At least 2 of Myl4, Kcnj3, Sln, Nr2f2
Ventricular-like	Myl2, Myl3, Hey2, Irx4, Fhl2	At least 1 of Myl2, Hey2, Irx4
Nodal-like candidate	Hcn4, Tbx3, Shox2, Isl1, Tbx18	At least 2 of Hcn4, Tbx3, Shox2
Conduction-associated CM candidate	Cntn2, Etv1, Gja5, Scn5a, Irx3	At least 2 of Cntn2, Etv1, Gja5
AV-like candidate	Tbx2, Tbx3, Bmp2, Msx2, Rspo3	At least 2 of Tbx2, Bmp2, Msx2

Fibroblast

Proposed identity	Positive marker panel	Required anchors
Postn-associated remodeling	Postn, Tnc, Fn1, Col1a1, Col3a1, Lox	At least 1 of Postn, Tnc, Lox
Gfpt2-enriched	Gfpt2, Pcsk6, Ebf1, Adamtsl3, Fbn1	At least 1 of Gfpt2
Kirrel3-enriched	Kirrel3, Bmper, Trps1, Abi3bp, Col5a2	At least 1 of Kirrel3
Col8a1-enriched	Col8a1, Col8a2, Cthrc1, Thbs2, Comp	At least 1 of Col8a1
Pi16/Dpt interstitial-like	Pi16, Dpt, Col14a1, Cd34, Col15a1	At least 1 of Pi16, Dpt
Lsamp-enriched candidate	Lsamp, Dpt, Col15a1, Fbn1	At least 1 of Lsamp

Endothelial

Proposed identity	Positive marker panel	Required anchors
Endocardial-like	Npr3, Nfatc1, Nrxn3, Cdh11, Alcam	At least 2 of Npr3, Nfatc1, Nrxn3
Capillary-like	Rgcc, Emcn, Car4, Cd36, Gpihbp1, Fabp4, Dach1	At least 2 of Rgcc, Emcn, Car4, Gpihbp1

Proposed identity	Positive marker panel	Required anchors
Arterial-like	Gja5, Efnb2, Sox17, Bmx, Hey1, Dll4	At least 2 of Gja5, Efnb2, Sox17, Bmx
Venous/venular-like	Nr2f2, Ephb4, Slc38a5, Ackr1, Vwf	At least 2 of Nr2f2, Ephb4, Ackr1
Lymphatic-like candidate	Prox1, Flt4, Lyve1, Pdpn, Ccl21a, Nrp2	At least 2 of Prox1, Flt4

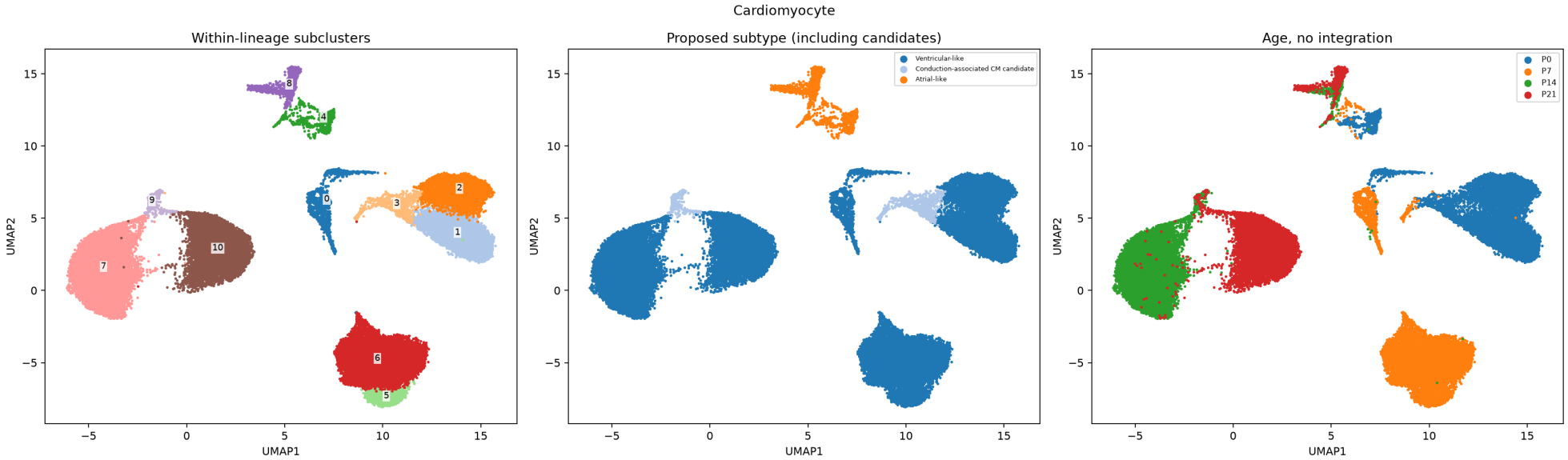
Immune

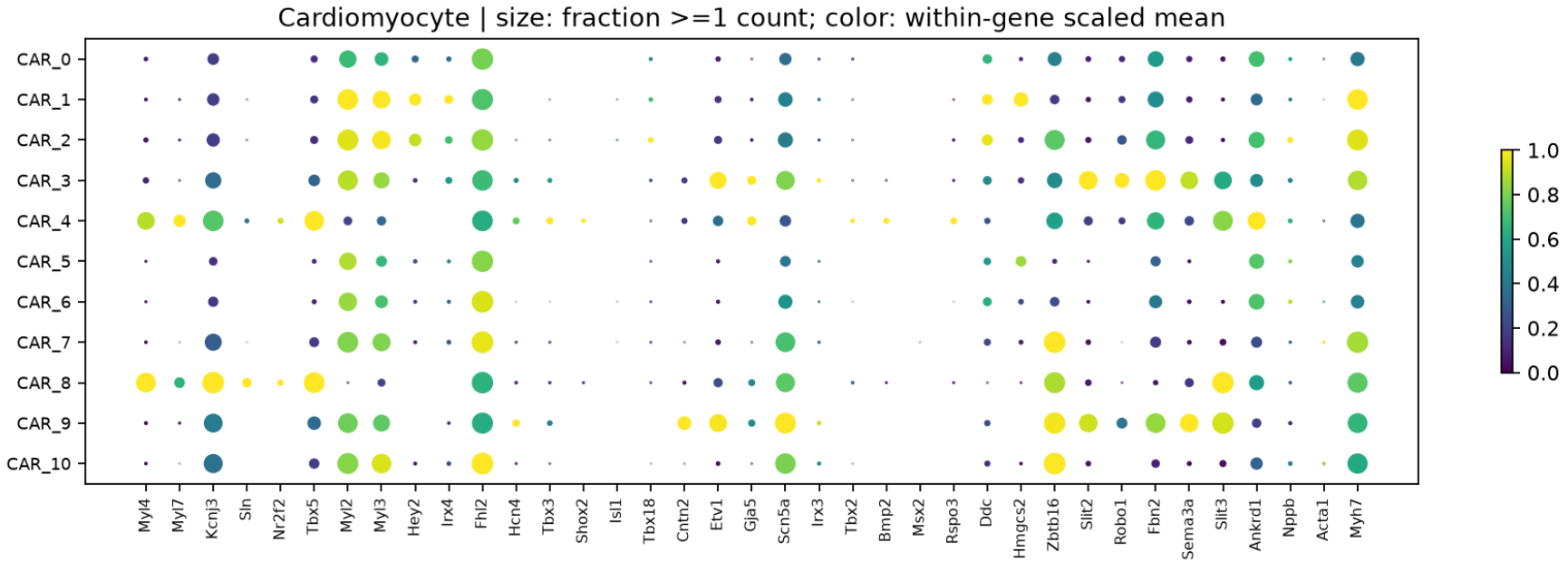
Proposed identity	Positive marker panel	Required anchors
Macrophage-like	Csf1r, Adgre1, Mertk, Mrc1, F13a1, C1qa	At least 2 of Csf1r, Adgre1, Mertk, C1qa
Monocyte-like	Ccr2, Ly6c2, Lyz2, S100a8, S100a9	At least 2 of Ccr2, Ly6c2, S100a8, S100a9
T-cell-like	Cd3d, Cd3e, Trbc1, Trbc2, Bcl11b	At least 2 of Cd3d, Cd3e, Trbc1, Trbc2
B-cell-like	Cd79a, Cd79b, Ms4a1, Pax5, Cd19	At least 2 of Cd79a, Cd79b, Ms4a1
NK-like	Nkg7, Ncr1, Klrd1, Prf1	At least 2 of Ncr1, Klrd1, Nkg7

Mural

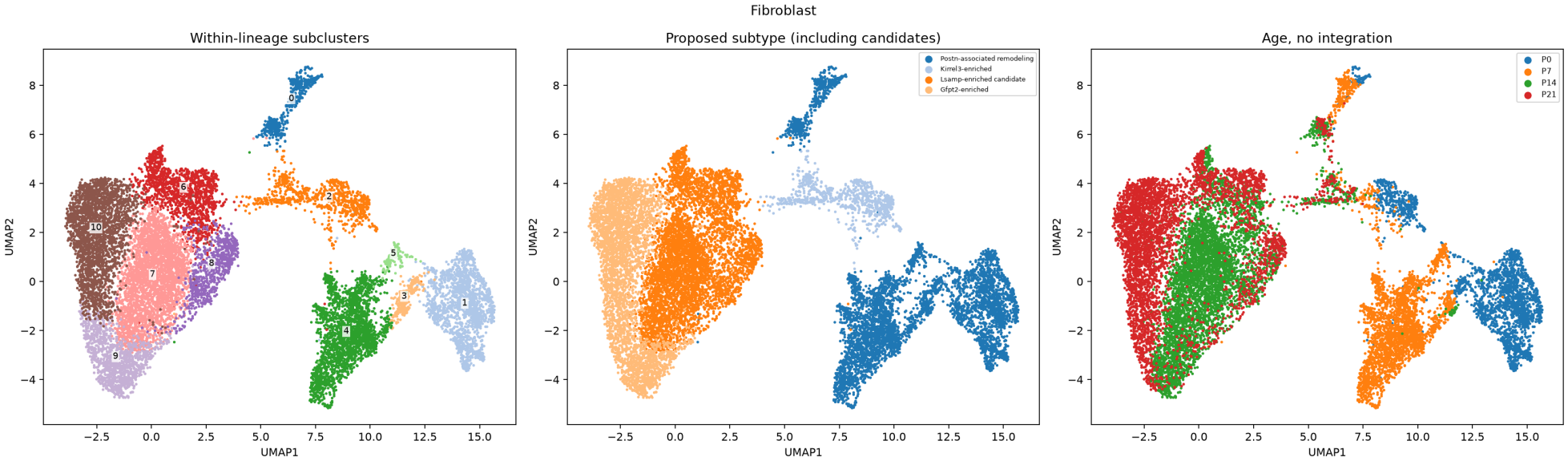
Proposed identity	Positive marker panel	Required anchors
Pericyte-like	Pdgfrb, Rgs5, Abcc9, Kcnj8, Cspg4, Notch3	At least 2 of Rgs5, Abcc9, Kcnj8, Cspg4
Smooth-muscle-like	Myh11, Acta2, Tagln, Cnn1, Lmod1	At least 2 of Myh11, Cnn1, Lmod1

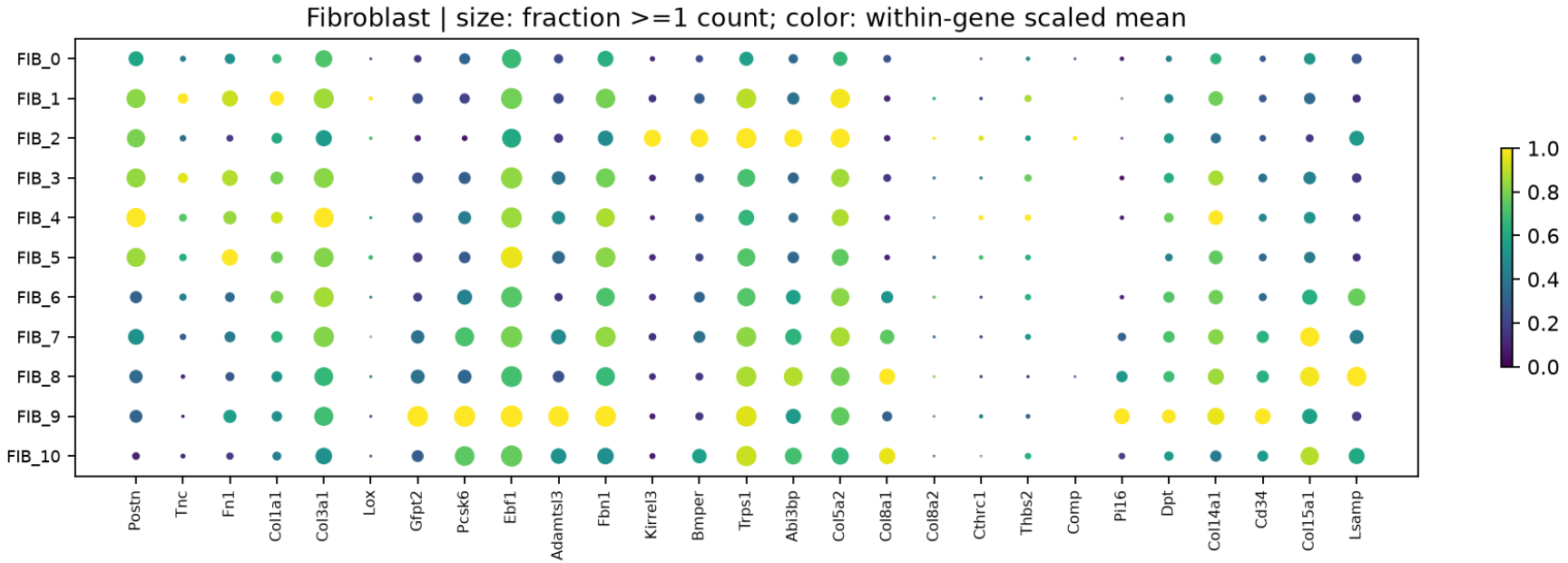
Cardiomyocyte subclustering



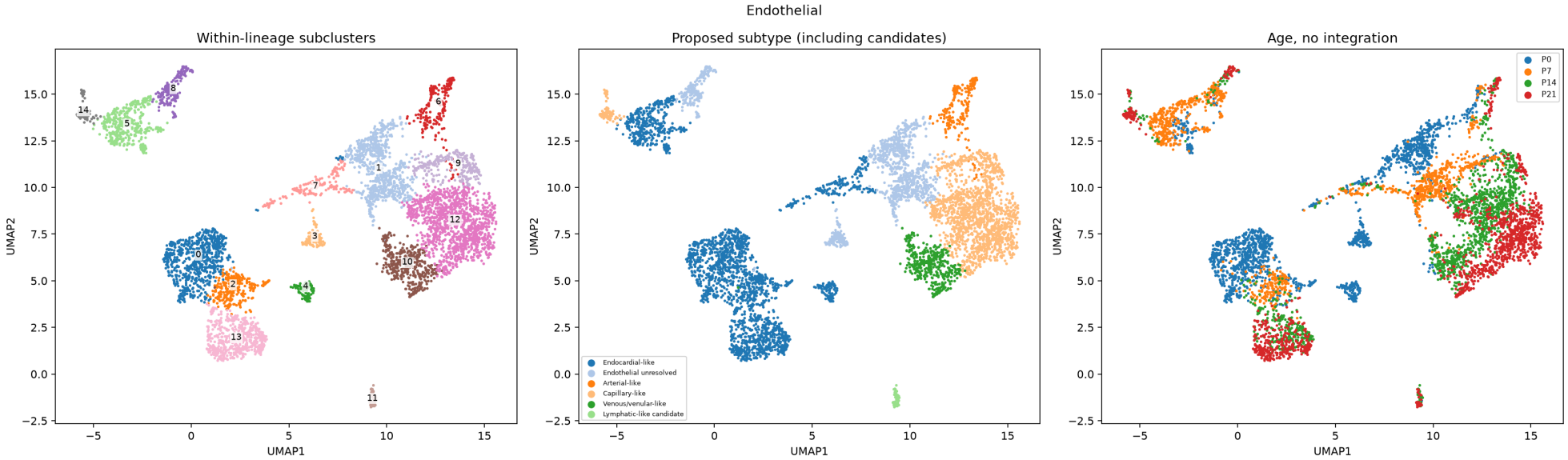


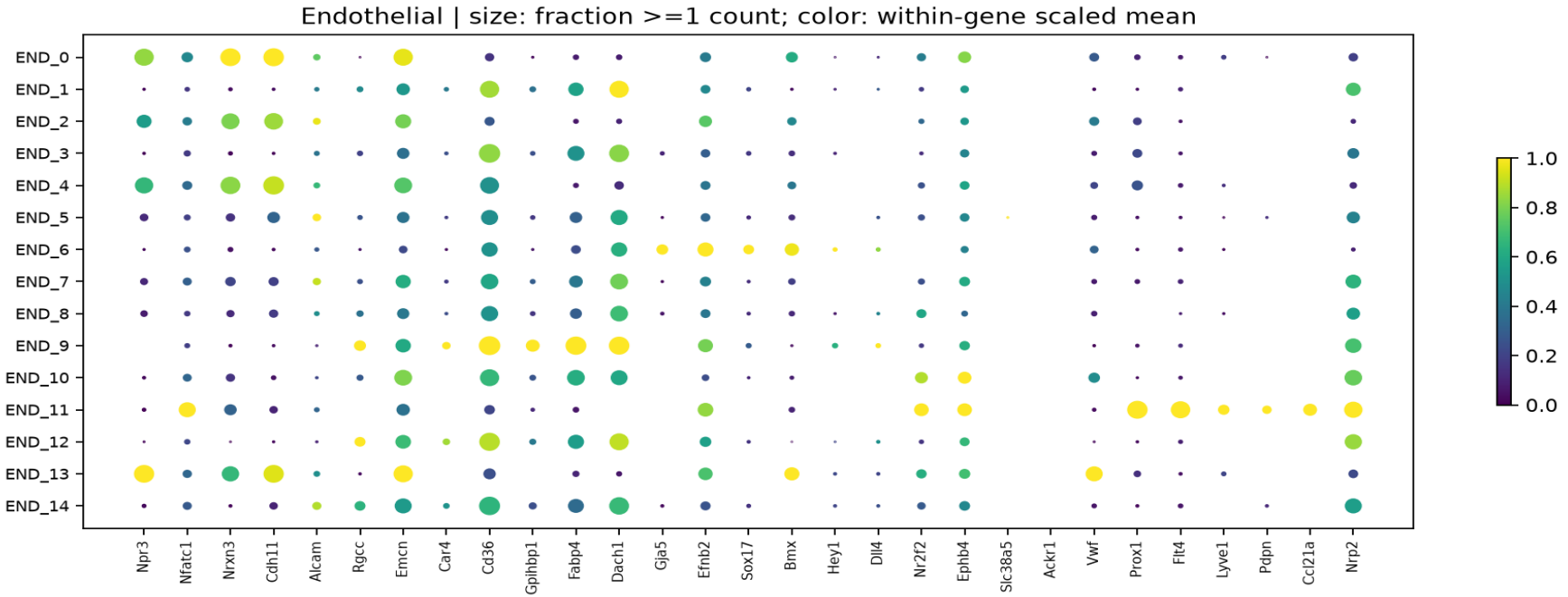
Fibroblast subclustering



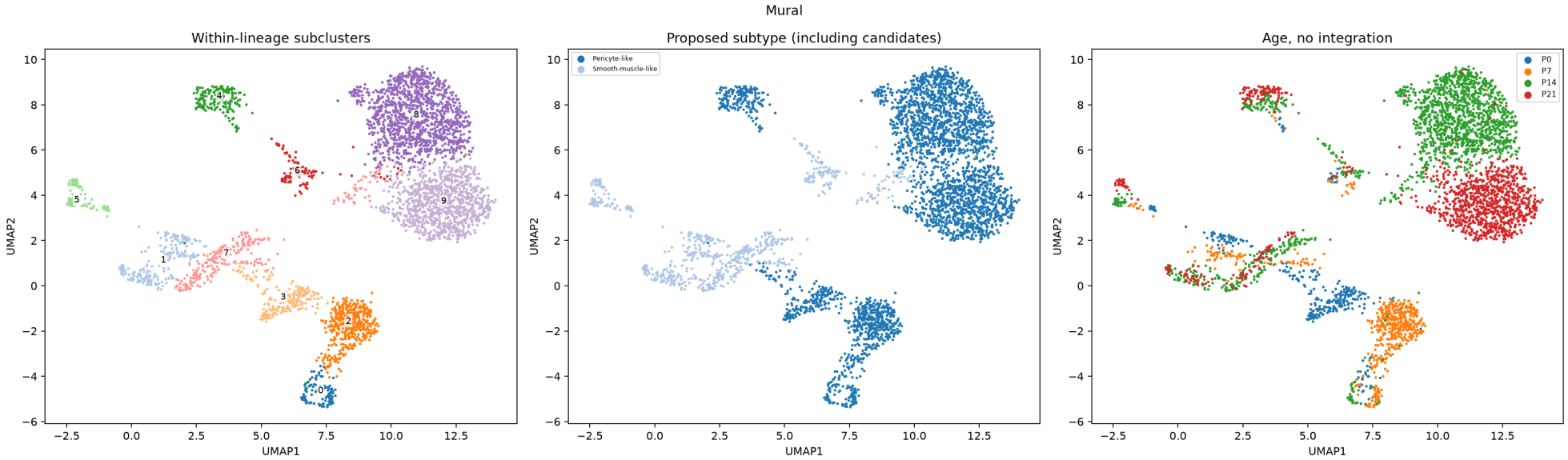


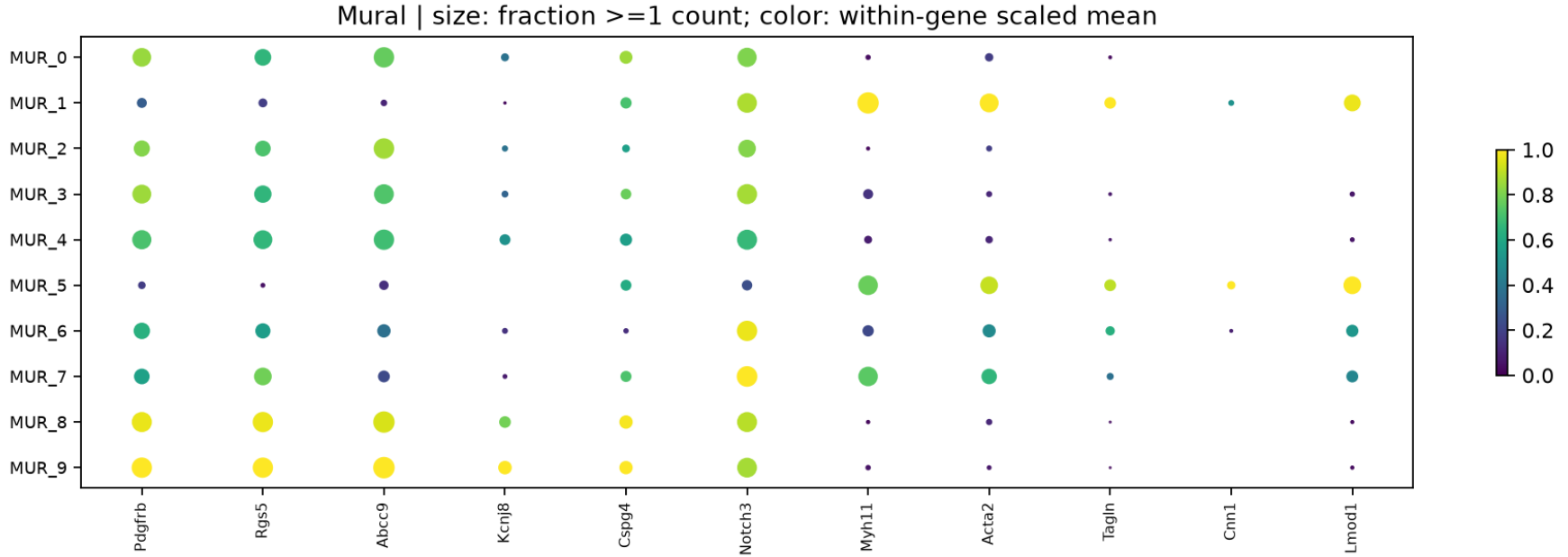
Endothelial subclustering




Endothelial | size: fraction >=1 count; color: within-gene scaled mean
END_0
END_1
END_2
END_3
END_4
END_5
END_6
END_7
END_8
END_9
END_10
END_11
END_12
END_13
END_14
Npr3
Nfatc1
Nrxn3
Cdh11
Alcam
Rgcc
Emcn
Car4
Cd36
Gpihbp1
Fabp4
Dach1
Gja5
Efnb2
Sox17
Bmx
Hey1
Dll4
Nr2f2
Ephb4
Slc38a5
Ackr1
Vwf
Prox1
Flt4
Lyve1
Pdpn
Ccl21a
Nrp2
1.0
0.8
0.6
0.4
0.2
0.0

Mural subclustering





Immune subclustering

